

AK4082 Model Linear Lanjut

Logistic Regression

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18 Mei 2024

```
library(dplyr)
library(pROC)
library(ggplot2)
library(caret)
# Membaca dan membersihkan data
data = read.csv("car.csv", header=TRUE)

data$numclaims = NULL
data$claimcst0 = NULL
data$X_OBSTAT_ = NULL

data$area = as.factor(data$area)
data$agecat = as.factor(data$agecat)
data$veh_age = as.factor(data$veh_age)
data$veh_body = as.factor(data$veh_body)
data$clm = as.factor(data$clm)
data$gender = as.factor(data$gender)
summary(data)

##      veh_value      exposure      clm      veh_body      veh_age
## Min.   : 0.000   Min.   :0.002738  0:63232  SEDAN   :22233  1:12257
## 1st Qu.: 1.010   1st Qu.:0.219028  1: 4624  HBACK   :18915  2:16587
## Median : 1.500   Median :0.446270      STNWG   :16261  3:20064
## Mean   : 1.777   Mean   :0.468651      UTE     : 4586  4:18948
## 3rd Qu.: 2.150   3rd Qu.:0.709103      TRUCK   : 1750
## Max.   :34.560   Max.   :0.999316      HDTOP   : 1579
##                                     (Other): 2532
##
## gender      area      agecat
## F:38603     A:16312    1: 5742
## M:29253     B:13341    2:12875
##                                     C:20540    3:15767
##                                     D: 8173    4:16189
##                                     E: 5912    5:10736
##                                     F: 3578    6: 6547
##

# Re-Level data Factor
data = data %>% mutate(veh_body = relevel(veh_body, ref = "SEDAN"))
data = data %>% mutate(veh_age = relevel(veh_age, ref = "3"))
data = data %>% mutate(gender = relevel(gender, ref = "F"))
data = data %>% mutate(area = relevel(area, ref = "C"))
data = data %>% mutate(agecat = relevel(agecat, ref = "4"))
```

Bagian 1

(Jong 2008, p.99–100). Occurrence of claim dimodelkan terhadap vehicle value dengan linear, quadratic, dan cubic model.

```
model1 = glm(clm ~ veh_value, family=binomial(link="logit"), data=data)
model2 = glm(clm ~ veh_value+I(veh_value^2), family=binomial(link="logit"), data=data)
model3 = glm(clm ~ veh_value+I(veh_value^2)+I(veh_value^3), family=binomial(link="logit"),
data=data)
Linear = c(round(as.numeric(summary(model1)$coefficients[,1]), 3),
NA, NA, model1$deviance, model1$rank, model1$aic)
```

```

Quadratic = c(round(as.numeric(summary(model2)$coefficients[,1]), 3),
              NA, model2$deviance, model2$rank, model2$aic)

Cubic      = c(round(as.numeric(summary(model3)$coefficients[,1]), 3),
              model3$deviance, model3$rank, model3$aic)

`Tabel 7.1` = data.frame(Linear, Quadratic, Cubic)
rownames(`Tabel 7.1`) = c("b0", "b1", "b2", "b3", "deviance", "p", "AIC")
`Tabel 7.1`

```

```

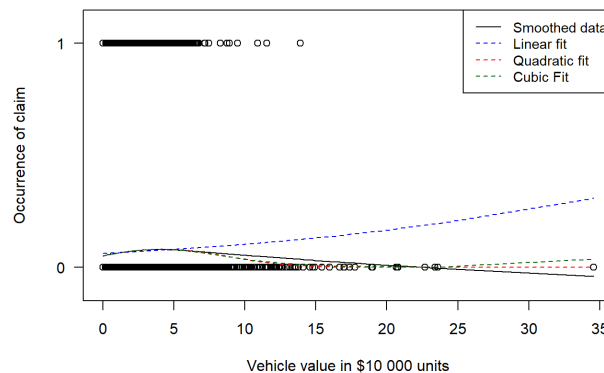
##           Linear Quadratic      Cubic
## b0        -2.716    -2.893    -2.925
## b1         0.055     0.220     0.261
## b2          NA    -0.026    -0.038
## b3          NA         NA     0.001
## deviance 33745.118 33712.921 33710.717
## p         2.000     3.000     4.000
## AIC      33749.118 33718.921 33718.717

```

```

plot(x=data$veh_value, y=data$clm, xlab="Vehicle value in $10 000 units", ylab="Occurrence of
claim", ylim=c(0.9,2.1), yaxt="n")
axis(side = 2, at=c(1,2), labels=c("0", "1"), las=2, mgp=c(3,1,0))
lines(smooth.spline(x=data$veh_value, y=data$clm), col="black")
lines(smooth.spline(x=data$veh_value, y=(model1$fitted.values+1)), col="blue", lty=2)
lines(smooth.spline(x=data$veh_value, y=(model2$fitted.values+1)), col="red", lty=2)
lines(smooth.spline(x=data$veh_value, y=(model3$fitted.values+1)), col="darkgreen", lty=2)
legend("topright", legend=c("Smoothed data", "Linear fit", "Quadratic fit", "Cubic Fit"),
      col=c("black", "blue", "red", "darkgreen"), lty=c(1,2,2,2), cex=0.9)

```



Bagian 2

(Jong 2008, p.103-105) Occurrence of claim dimodelkan terhadap driver's age, gender, area, vehicle body, vehicle's age, dan vehicle value yang di-banding.

```

# Band scheme (Tabel 7.2)
data2 = data
data2$veh_value = ifelse(data2$veh_value <= 2.5, "<=25",
                        ifelse(data2$veh_value <= 5, "25-50",
                              ifelse(data2$veh_value <= 7.5, "50-75",
                                    ifelse(data2$veh_value <= 10, "75-100",
                                          ifelse(data2$veh_value <= 12.5, "100-125", ">125")))))

data2$veh_value = as.factor(data2$veh_value)
data2 = data2 %>% mutate(veh_value = relevel(veh_value, ref = "<=25"))

summary(data2)

```

```

##   veh_value   exposure    clm   veh_body   veh_age
## <=25   :54971   Min.    :0.002738 0:63232  SEDAN :22233  3:20064
## >125    :   33   1st Qu.:0.219028 1: 4624  HBACK :18915  1:12257
## 100-125:   44   Median :0.446270          STNWX :16261  2:16587

```

```
## 25-50 :11439 Mean :0.468651 UTE : 4586 4:18948
## 50-75 : 1265 3rd Qu.:0.709103 TRUCK : 1750
## 75-100 : 104 Max. :0.999316 HDTOP : 1579
## (Other): 2532
##
## gender area agecat
## F:38603 C:20540 4:16189
## M:29253 A:16312 1: 5742
## B:13341 2:12875
## D: 8173 3:15767
## E: 5912 5:10736
## F: 3578 6: 6547
##
```

Model Tanpa Eksposur

```
model1 = step(glm(c1m ~ veh_value+veh_body+veh_age+gender+area+agecat,
family=binomial(link="logit"), data=data2), direction="both", trace=FALSE)
summary(model1)
```

```
##
## Call:
## glm(formula = c1m ~ veh_body + veh_age + area + agecat, family = binomial(link = "logit"),
## data = data2)
##
## Deviance Residuals:
## Min 1Q Median 3Q Max
## -0.7609 -0.3921 -0.3696 -0.3447 2.6954
##
## Coefficients:
## Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.620132 0.049436 -53.000 < 2e-16 ***
## veh_bodyBUS 1.147048 0.372303 3.081 0.002063 **
## veh_bodyCONVT -0.666933 0.589106 -1.132 0.257588
## veh_bodyCOUPE 0.296104 0.130508 2.269 0.023277 *
## veh_bodyHBACK -0.030735 0.040036 -0.768 0.442668
## veh_bodyHDTOP 0.215956 0.096381 2.241 0.025049 *
## veh_bodyMCARA 0.655566 0.285292 2.298 0.021569 *
## veh_bodyMIBUS -0.085321 0.160622 -0.531 0.595284
## veh_bodyPANVN 0.202005 0.135714 1.488 0.136628
## veh_bodyRDSTR 0.046284 0.736181 0.063 0.949869
## veh_bodySTNWG 0.069530 0.041733 1.666 0.095700 .
## veh_bodyTRUCK 0.011346 0.099682 0.114 0.909375
## veh_bodyUTE -0.193280 0.070540 -2.740 0.006144 **
## veh_age1 -0.002545 0.045992 -0.055 0.955863
## veh_age2 0.124189 0.040769 3.046 0.002318 **
## veh_age4 -0.093283 0.041458 -2.250 0.024445 *
## areaA -0.036845 0.041877 -0.880 0.378944
## areaB 0.060444 0.043396 1.393 0.163663
## areaD -0.122776 0.054439 -2.255 0.024115 *
## areaE -0.046894 0.060212 -0.779 0.436092
## areaF 0.078805 0.070177 1.123 0.261458
## agecat1 0.257382 0.056756 4.535 5.76e-06 ***
## agecat2 0.058293 0.046389 1.257 0.208892
## agecat3 0.031920 0.044212 0.722 0.470311
## agecat5 -0.188547 0.052043 -3.623 0.000291 ***
## agecat6 -0.200331 0.062591 -3.201 0.001371 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 33767 on 67855 degrees of freedom
## Residual deviance: 33615 on 67830 degrees of freedom
## AIC: 33667
##
## Number of Fisher Scoring iterations: 5
```

Model dengan Exposure

```
model2 = step(glm(c1m ~ veh_value+veh_body+veh_age+gender+area+agecat, offset=exposure,
family=binomial(link="logit"), data=data2), direction="both", trace=FALSE)
```

```
summary(model2)
```

```
##
## Call:
## glm(formula = clm ~ veh_body + veh_age + area + agecat, family = binomial(link = "logit"),
##      data = data2, offset = exposure)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -0.8078  -0.4072  -0.3553  -0.3126   2.6935
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -3.130580   0.049542 -63.190 < 2e-16 ***
## veh_bodyBUS    1.102183   0.373857   2.948  0.00320 **
## veh_bodyCONVT  -0.619063   0.590364  -1.049  0.29436
## veh_bodyCOUPE  0.360406   0.130930   2.753  0.00591 **
## veh_bodyHBACK  -0.032627   0.040159  -0.812  0.41655
## veh_bodyHDTOP   0.189774   0.096722   1.962  0.04976 *
## veh_bodyMCARA   0.663660   0.286775   2.314  0.02066 *
## veh_bodyMIBUS  -0.054149   0.161021  -0.336  0.73665
## veh_bodyPANVN   0.124632   0.136218   0.915  0.36022
## veh_bodyRDSTR   0.062921   0.738118   0.085  0.93207
## veh_bodySTNWG   0.068543   0.041842   1.638  0.10139
## veh_bodyTRUCK  -0.006011   0.099999  -0.060  0.95207
## veh_bodyUTE    -0.185473   0.070715  -2.623  0.00872 **
## veh_age1        0.037289   0.046127   0.808  0.41885
## veh_age2        0.123158   0.040902   3.011  0.00260 **
## veh_age4       -0.094435   0.041569  -2.272  0.02310 *
## areaA          -0.036081   0.041996  -0.859  0.39026
## areaB           0.056506   0.043523   1.298  0.19418
## areaD          -0.122707   0.054593  -2.248  0.02460 *
## areaE          -0.047721   0.060393  -0.790  0.42942
## areaF           0.059730   0.070404   0.848  0.39622
## agecat1         0.274229   0.056936   4.816 1.46e-06 ***
## agecat2         0.071686   0.046524   1.541  0.12336
## agecat3         0.033045   0.044337   0.745  0.45607
## agecat5        -0.201731   0.052172  -3.867  0.00011 ***
## agecat6        -0.206425   0.062752  -3.290  0.00100 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 32813  on 67855  degrees of freedom
## Residual deviance: 32656  on 67830  degrees of freedom
## AIC: 32708
##
## Number of Fisher Scoring iterations: 5
```

```
# Deviance Analysis
anova(model1, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: clm
##
## Terms added sequentially (first to last)
##
##
##              Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
## NULL                67855      33767
## veh_body  12      39.547      67843      33727 8.547e-05 ***
## veh_age   3      28.798      67840      33698 2.469e-06 ***
## area      5      16.861      67835      33682  0.00477 **
## agecat    5      66.572      67830      33615 5.286e-13 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model2, test="Chisq")
```

```
## Analysis of Deviance Table
```

```
##
```

```
## Model: binomial, link: logit
```

```
##
```

```
## Response: c1m
```

```
##
```

```
## Terms added sequentially (first to last)
```

```
##
```

```
##
```

```
##          Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
```

```
## NULL                67855      32813
```

```
## veh_body 12    36.698    67843    32776 0.0002501 ***
```

```
## veh_age   3    29.193    67840    32747 2.040e-06 ***
```

```
## area      5    15.385    67835    32731 0.0088390 **
```

```
## agecat    5    75.268    67830    32656 8.179e-15 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Selang Kepercayaan Koefisien
```

```
# Note: di sini tidak di run karena runningnya lama
```

```
# confint(model1, level=0.95)
```

```
# confint(model2, level=0.95)
```

ROC Curve

Kurva ROC (Receiver Operating Characteristic) adalah plot dua dimensi yang menggambarkan seberapa baik sistem classifier bekerja saat nilai threshold berubah selama rentang dari variabel prediktor.

```
# Membangun ROC curve
```

```
pred1 = predict(model1, type="response")
```

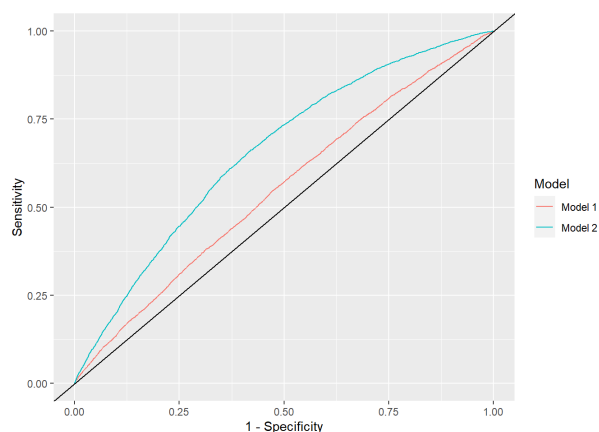
```
pred2 = predict(model2, type="response")
```

```
roccurve1 = roc(data2$c1m ~ pred1)
```

```
roccurve2 = roc(data2$c1m ~ pred2)
```

```
roclist = list("Model 1"=roccurve1, "Model 2"=roccurve2)
```

```
ggroc(roclist,aes="colour",legacy.axes = TRUE) + geom_abline(intercept = 0,slope = 1) +  
labs(x = "1 - Specificity", y = "Sensitivity", colour = "Model")
```



```
(roccurve1$auc)

## Area under the curve: 0.551

(roccurve2$auc)

## Area under the curve: 0.6573
```

Area Under Curve (AUC) adalah luas dari daerah di bawah kurva ROC. AUC adalah suatu ukuran yang dapat merepresentasikan seberapa baik model dalam membedakan antar kelas klasifikasi. Secara umum, rule of thumb dalam menginterpretasi AUC ada pada tabel di bawah (Hosmer Jr dkk., 2013).

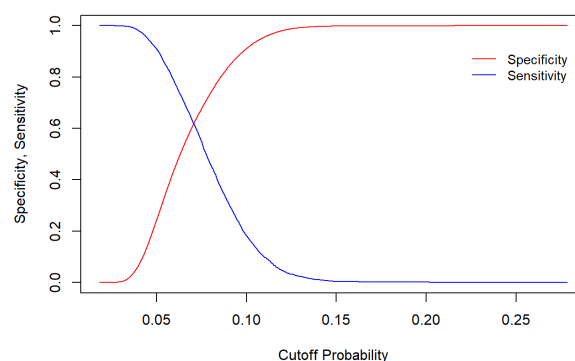
Nilai AUC	Interpretasi
$AUC = 0.5$	Tidak ada perbedaan, misal melempar koin acak
$0.5 < AUC < 0.7$	Pembedaan yang buruk
$0.7 \leq AUC < 0.8$	Pembedaan yang dapat diterima
$0.8 \leq AUC < 0.9$	Pembedaan yang baik
$AUC \geq 0.9$	Pembedaan yang sangat baik

Tabel Kontingensi

Sebuah confusion matrix (juga dikenal sebagai matriks error) adalah tabel kontingensi yang digunakan untuk menggambarkan kinerja pengklasifikasi/sistem klasifikasi, ketika kebenaran diketahui.

```
library(ROCR)

pred_test = predict(model2, type=c("response"))
pred      = prediction(pred_test, data2$clm)
evalf     = performance(pred, "f")
sens      = performance(pred, "sens")
spec      = performance(pred, "spec")
akurasi   = performance(pred, "acc")
prec      = performance(pred, "prec")
SeSAccPrec = cbind(Cutoff=sens@x.values[[1]], Se=sens@y.values[[1]], Sp=spec@y.values[[1]],
                  Acc=akurasi@y.values[[1]], Prec=prec@y.values[[1]], F1=evalf@y.values[[1]])
SeSAccPrec = data.frame(SeSAccPrec)
plot(x = SeSAccPrec$Cutoff, y=SeSAccPrec$Sp, type="l", col="red", xlab="Cutoff Probability",
     ylab="Specificity, Sensitivity")
lines(x=SeSAccPrec$Cutoff, y=SeSAccPrec$Se, col="blue")
legend("topright", legend=c("Specificity", "Sensitivity"), col=c("Red", "Blue"), cex=0.9,
     lty=1, box.lty=0, inset=c(0.025, 0.1))
```



```
coords(roccurve2, x="best", input="threshold", best.method="youden")
```

```
## threshold specificity sensitivity
## 1 0.06854489 0.5859533 0.6559256
```

```
# Menyusun tabel kontingensi dengan threshold yang menyeimbangkan sensitivitas dan spesifisitas
```

```
pred = factor(ifelse(model2$fitted.values < 0.06854489, 0, 1))
mat1 = confusionMatrix(pred, data$c1m, positive = "1")
mat1
```

```
## Confusion Matrix and Statistics
```

```
##
##           Reference
## Prediction    0    1
##           0 37051 1591
##           1 26181 3033
##
##           Accuracy : 0.5907
##           95% CI : (0.587, 0.5944)
##           No Information Rate : 0.9319
##           P-Value [Acc > NIR] : 1
##
##           Kappa : 0.0698
##
## Mcnemar's Test P-Value : <2e-16
##
##           Sensitivity : 0.65593
##           Specificity : 0.58595
##           Pos Pred Value : 0.10382
##           Neg Pred Value : 0.95883
##           Prevalence : 0.06814
##           Detection Rate : 0.04470
##           Detection Prevalence : 0.43053
##           Balanced Accuracy : 0.62094
##
##           'Positive' Class : 1
##
```

	Reference	
Predicted	No Event	Event
No Event	D	C
Event	B	A

Formula yang digunakan:

$$sensitivity = \frac{A}{A + C}$$

$$specificity = \frac{D}{B + D}$$

$$prevalence = \frac{A + C}{A + B + C + D}$$

$$PPV = \frac{sensitivity \times prevalence}{((sensitivity \times prevalence) + (1 - specificity) \times (1 - prevalence))}$$

$$NPV = \frac{specificity \times (1 - prevalence)}{((1 - sensitivity) \times prevalence + specificity \times (1 - prevalence))}$$

$$detection\ rate = \frac{A}{A + B + C + D}$$

$$detection\ prevalence = \frac{A + B}{A + B + C + D}$$

$$balanced\ accuracy = (sensitivity + specificity)/2$$

- Accuracy = $(A + D)/(A + B + C + D)$ menunjukkan tingkat kecocokan prediksi dengan data asli, dalam kasus ini, 59.07% dari prediksi cocok dengan data asli dengan menggunakan threshold 0.06854489.
- Sensitivity adalah proporsi dari True Positive (Hasil prediksi '1', dan memang data aslinya '1') dengan jumlah kategori '1' pada data. Berarti, 65.593% dari kategori '1' pada data berhasil diprediksi dengan benar.
- Specificity adalah proporsi dari True Negative (Hasil prediksi '0', dan memang data aslinya '0') dengan jumlah kategori '0' pada data. Berarti, 58.595% dari kategori '0' pada data berhasil diprediksi dengan benar.
- Prevalence adalah proporsi jumlah kategori '1' pada data dengan total data. Maka 6.814% dari data ini merupakan kategori '1'.
- Positive Predicted Value adalah persentase dari True Positive dari seluruh hasil prediksi yang positif. Berarti, dari seluruh prediksi yang merupakan kategori '1', terdapat 10.382% yang benar bernilai '1' pada data.
- Negative Predicted Value adalah persentase dari True Negative dari seluruh hasil prediksi yang negatif. Berarti, dari seluruh prediksi yang merupakan kategori '0', terdapat 95.883% yang benar bernilai '0' pada data.

Prediksi

```
prob      = predict(model2, data, type='response')
predtest = factor(ifelse(prob < 0.06854489, 0, 1))
head(data.frame(FittedValue=predtest,data))
```

##	FittedValue	veh_value	exposure	clm	veh_body	veh_age	gender	area	agecat
## 1	0	1.06	0.3039014	0	HBACK	3	F	C	2
## 2	1	1.03	0.6488706	0	HBACK	2	F	A	4
## 3	1	3.26	0.5694730	0	UTE	2	F	E	2
## 4	0	4.14	0.3175907	0	STNWG	2	F	D	2
## 5	1	0.72	0.6488706	0	HBACK	4	F	C	2
## 6	1	2.01	0.8542094	0	HDTOP	3	M	C	4